Women and other Minorities in IISERs and IISc

A data study of enrolment, 2017-18 to 2024-25 · iGEM IISER Kolkata 2026 · Inclusivity ·

Background

Women are about 35% of higher-education STEM students and about 28% of researchers worldwide [1]. In India they are under 15% of researchers [2]. The IISERs are national institutes for science education and research, so who enrols there shapes who becomes a researcher.

Aims

To describe, with public data and reproducible code, how many women are enrolled in the IISERs, how this has changed since 2017-18, and what the data can and cannot tell us about why.

Results

- Women are 27% of enrolled five-year (BS-MS) students at IISER Kolkata and 28% at IISER Pune, but 52% at Tirupati and 52% at TVM, a gap of about 24 points.
- IISER Kolkata enrolls the fewest women of the seven IISERs (286), about a third of TVM's 793.
- Since 2017-18 there is no general decline: the share rose at Mohali (+7.4 points), TVM (+4.4) and Kolkata (+3.1), and fell slightly at Pune (−2.2).
- Seat expansion did not help evenly. Of the net growth in enrolment, 77% went to women at Mohali, 57% at TVM, 50% at Kolkata and 23% at Pune.
- A larger share of students from the home state does not explain the differences ($r = -0.06$, $n = 7$).
- At Pune and Kolkata, postgraduate programmes have a higher female share than the five-year programme (40% against 28%, and 33% against 27%), which points to entry into the undergraduate programme as one place to look.

Introduction

Who studies science decides who does science. Worldwide, women are about 35% of students in higher-education STEM and about 28% of researchers [1]. In India, UNESCO Institute for Statistics figures put women at under 15% of researchers [2]. This is not only a question of fairness. A large study of US doctoral theses found that women and other underrepresented groups introduced new combinations of ideas at higher rates than their peers, yet their work was taken up less and rewarded less [3]. A research system that loses some people early also loses some of its ideas.

For our Inclusivity work, we wanted to look at our own surroundings with numbers instead of impressions. Every year, Indian institutes submit enrolment data to the National Institutional Ranking Framework (NIRF) [4]. The seven IISERs (Kolkata, Pune, Mohali, Bhopal, Thiruvananthapuram, Tirupati and Berhampur) are directly comparable. They run the same five-year integrated BS-MS programme, recruit

nationally, and report in the same format. We asked three questions. How many women are enrolled? Is that share changing? And what would we need to know to explain it?

Note: We want to be clear about what this study is. It describes enrollment. It does not show causes, and it cannot speak for individual students. Where we offer explanations, we mark them as questions to test.

Our dataset

We collected the public NIRF "Overall" data submissions of the seven IISERs, and of IISc Bangalore for reference. That is 44 PDFs. From each we read the table "Total Actual Student Strength", which lists male and female students for each degree programme, together with how many come from the home state, other states and abroad, and how many belong to the economically backward and socially challenged categories.

A script extracted the numbers into one table of 79 rows. In every row, male plus female equals the total, and home-state plus other-state plus abroad equals the total.

The main comparison uses the five-year BS-MS programme ("UG 5-yr"), which all seven IISERs offer. Bhopal also reports a four-year programme, and IISc a four-year undergraduate programme. These are not comparable and are discussed separately.

DOWNLOAD THE DATABASE

IISER_enrolment_data.xlsx (all parsed rows, trend tables, intake) and parse_nirf.py (the extraction script)

Table 1. Files are attached in the wiki for reference.. NIRF year Y describes the academic year (Y−2) to (Y−1).

Institute	Files	Years	Data for
Kolkata	8	2019-26	2017-18 to 24-25
Pune	8	2019-26	2017-18 to 24-25
Mohali	7	2019-25	2017-18 to 23-24
TVM	7	2020-26	2018-19 to 24-25
Berhampur	3	2024-26	2022-23 to 24-25
Tirupati	2	2025-26	2023-24 to 24-25
Bhopal	1	2026	2024-25
IISc (reference)	8	2018, 20-26	2016-17 to 24-25

How to read the figures. Each figure is the share of women among all students currently enrolled across the years of a programme, not among new admissions, so changes appear slowly. Gender is recorded only as male or female, so these data cannot show non-binary or gender-diverse students. The institutes submit the numbers themselves and they are not independently audited. Because the tables count every enrolled student, they are not a sample and we do not report confidence intervals.

Main findings

1. Large and persistent differences between institutes

In the latest year available, the female share of the five-year programme ranges from 27.2% at Kolkata to 51.8% at Tirupati. Kolkata and Pune (27.6%) sit together at the bottom, about 24 to 25 points below Tirupati and TVM (51.6%). Berhampur, Mohali and Bhopal lie between 39% and 44%.

In 2023-24, the six institutes with data that year together enrolled 41.5% women. Kolkata, at 27.2% in 2024-25, was about 15 points below that.

The order has been stable. Kolkata and Pune have been the two lowest at almost every point since 2017-18, and Mohali has stayed above them.

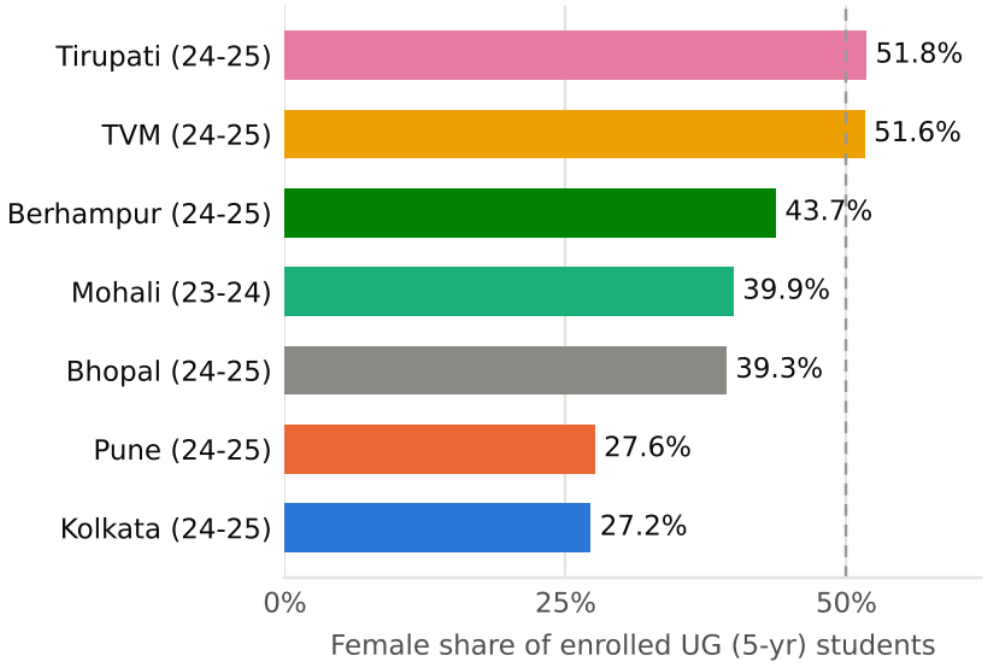


Figure 1: Female share of enrolled UG five-year students, latest year. Dashed line: 50%. Mohali's latest file covers 2023-24.

Counting people shows the gap differently. IISER Kolkata has the fewest women of the seven institutes (286), although it enrols 1,050 students against Pune's 1,232 and Mohali's 1,139 (2023-24). Mohali has 455 women. TVM has 793 women, 2.8 times as many, with only about 1.5 times as many students (1,536).

Bhopal's four-year undergraduate programme has a lower share than any five-year programme, at 19.8% (114 of 575 students in 2024-25). For reference, IISc's four-year undergraduate programme has stayed between about 12% and 17% since 2016-17.

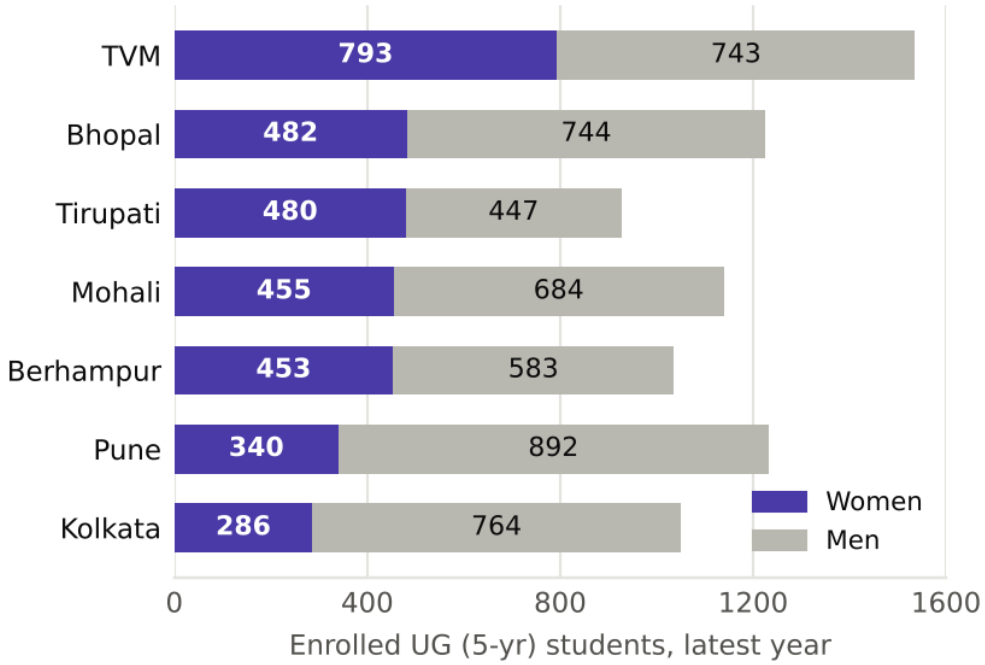


Figure 2: Number of women and men enrolled in the UG five-year programme, latest year. Violet: women. Grey: men.

2. Slow and uneven change since 2017-18

Over the years we have a mixed picture and not a general decline.

- **Mohali:** 32.5% to 39.9% (2017-18 to 2023-24), the largest rise.
- **TVM:** 47.2% in 2018-19, a peak of 55.3% in 2022-23, then 51.6% in 2024-25.
- **Kolkata:** 24.2% to 27.2%, mostly from a step up in 2020-21 and roughly flat since.
- **Pune:** 29.8% in 2017-18, a low of 26.2% in 2019-20, and 27.6% in 2024-25, with no clear direction.
- **Berhampur and Tirupati:** three and two years of data, with small changes.

For the three institutes with data from 2017-18 (Kolkata, Pune, Mohali), the combined female share went from 28.8% to 32.0% by 2023-24, a rise of 3.1 points.

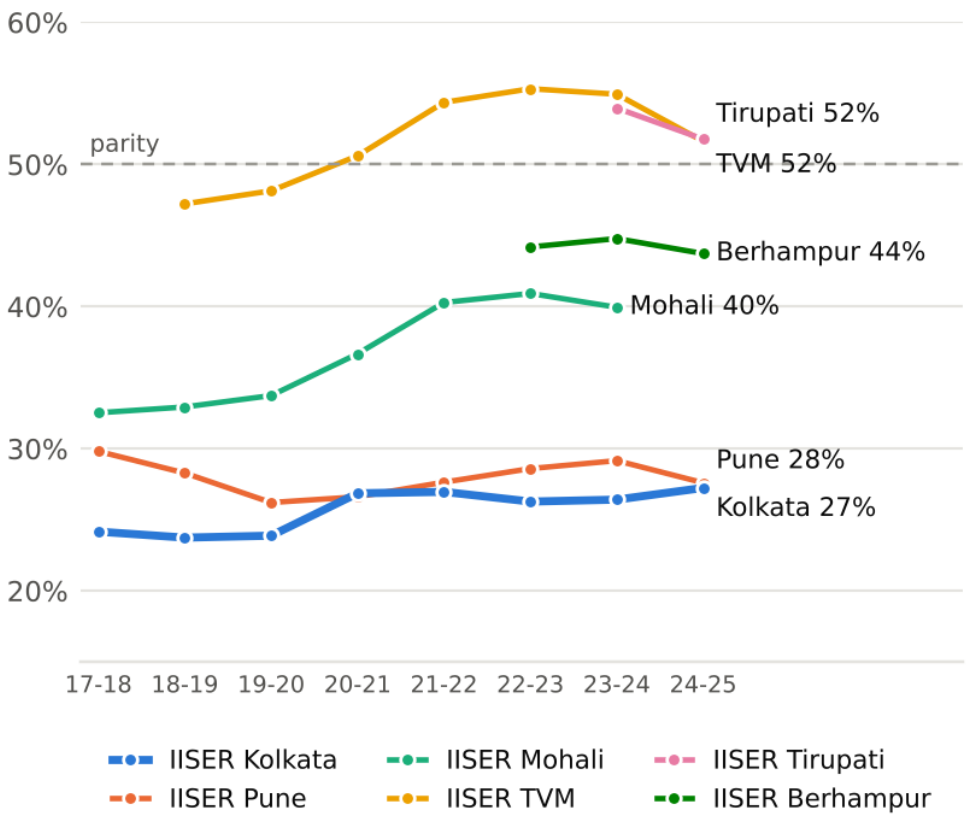


Figure 3: Female share of enrolled UG five-year students over time. Lines start where data exist. TVM starts in 2018-19, Berhampur in 2022-23, Tirupati in 2023-24; Mohali ends in 2023-24.

Table 2. Change in female share (UG five-year programme), first to last year available.

Institute	Period	Start (%)	End (%)	Change (points)	Fitted trend (points per year)
IISER Kolkata	2017-18 to 2024-25	24.2	27.2	+3.1	+0.5
IISER Pune	2017-18 to 2024-25	29.8	27.6	−2.2	about 0
IISER Mohali	2017-18 to 2023-24	32.5	39.9	+7.4	+1.6
IISER TVM	2018-19 to 2024-25	47.2	51.6	+4.4	+1.1
IISER Berhampur	2022-23 to 2024-25	44.2	43.7	−0.5	−0.2
IISER Tirupati	2023-24 to 2024-25	54.0	51.8	−2.2	too few years

As a rough illustration only, Kolkata's fitted trend is about 0.5 points per year. At that pace, reaching 50% would take roughly 45 to 55 years. This extends a short trend and is not a forecast.

3. The seats grew. Who got the growth?

All the institutes have added seats. Kolkata, Pune and Mohali each started at 200 sanctioned places in 2013-14 and now have 250, 248 and 244 (Mohali in 2023-24). TVM went from 200 to 320. Tirupati went from 150 in 2019-20 to 300. Berhampur, a newer institute, went from 106 to 180, and Bhopal from 252 to 290.

Kolkata, Pune and Mohali expanded by similar proportions (about a quarter) from 200 seats, yet their female shares moved by +3.1, −2.2 and +7.4 points. So expansion in itself did not raise the share. What mattered was who filled the new places.

Sanctioned intake is the approved capacity, not the number admitted, and the figure for a year can be revised in later reports. We use the most recent report that mentions each year.

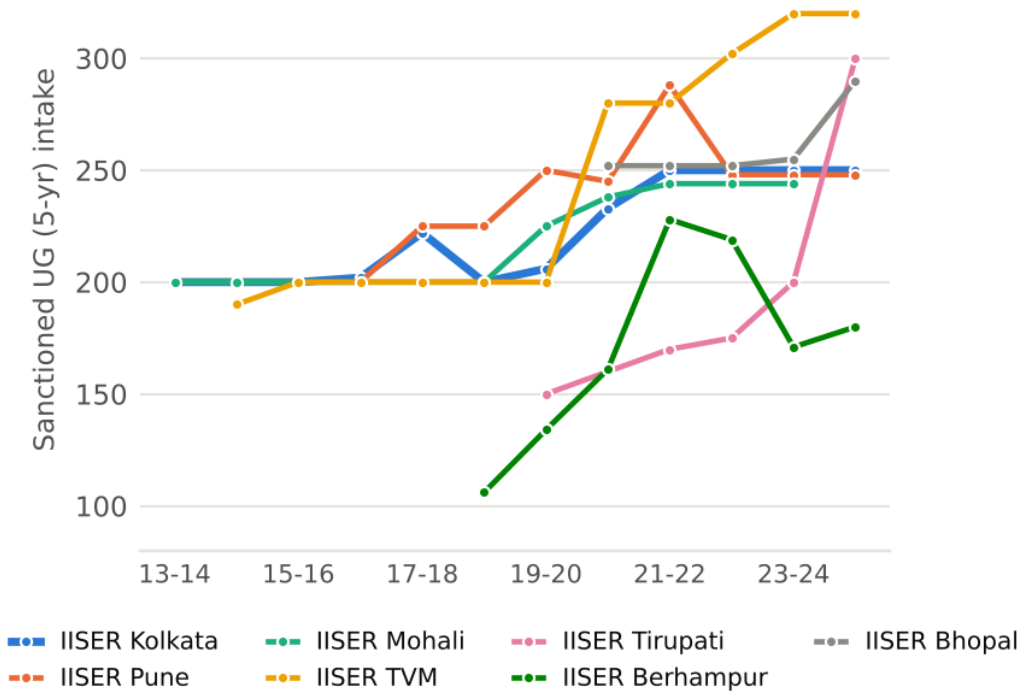


Figure 4: Sanctioned intake for the UG five-year programme. Seats approved per year, from NIRF intake tables.

Counting the net change in enrolment shows where the extra students went. At Pune, enrolment grew by 294 men and 86 women, so only 23% of the growth was women. That is below Pune's starting share of 30%, which is why its share did not rise. At Mohali, 147 women and 45 men were added (77% women), and the share rose by about 7 points.

Kolkata is different again. About half of its growth was women (63 women and 64 men), far above its starting share, but its total enrollment grew by only 127 students (923 to 1,050), so the share moved only about 3 points.

Because we count all enrolled students across five cohorts, these figures cannot tell us the gender mix of each year's new entrants. The institutes hold that information.

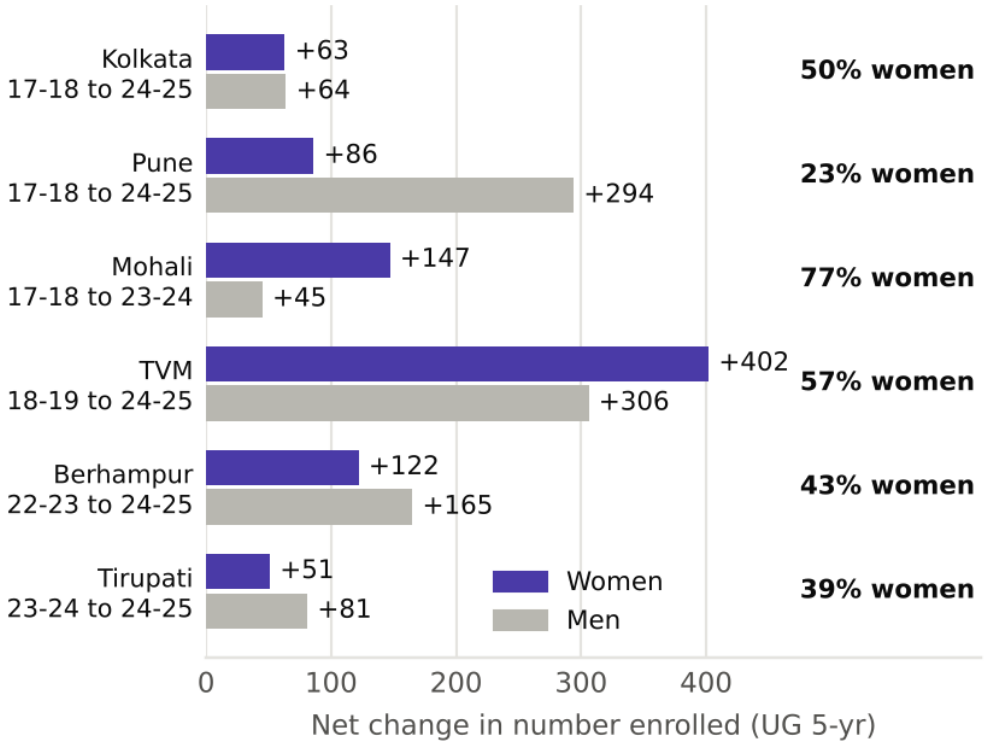


Figure 5: Net change in enrolment, women and men. Bold: share of the net growth that was women. Periods differ by institute.

4. Undergraduate or postgraduate?

Five institutes also report postgraduate students. At Pune the female share of all postgraduate programmes is 40%, against 28% in the five-year programme. At Kolkata, it is 33% against 27%. At TVM the two are about equal (51% and 52%). At Tirupati and Bhopal, the postgraduate share is lower than the five-year share.

One reading is that at Pune and Kolkata the narrowest point is entry into the undergraduate programme, not later stages. This is for two reasons. The postgraduate groups are small (73 and 139 students). And they recruit through different routes, including students from other institutions, so they are not a continuation of the same cohort.

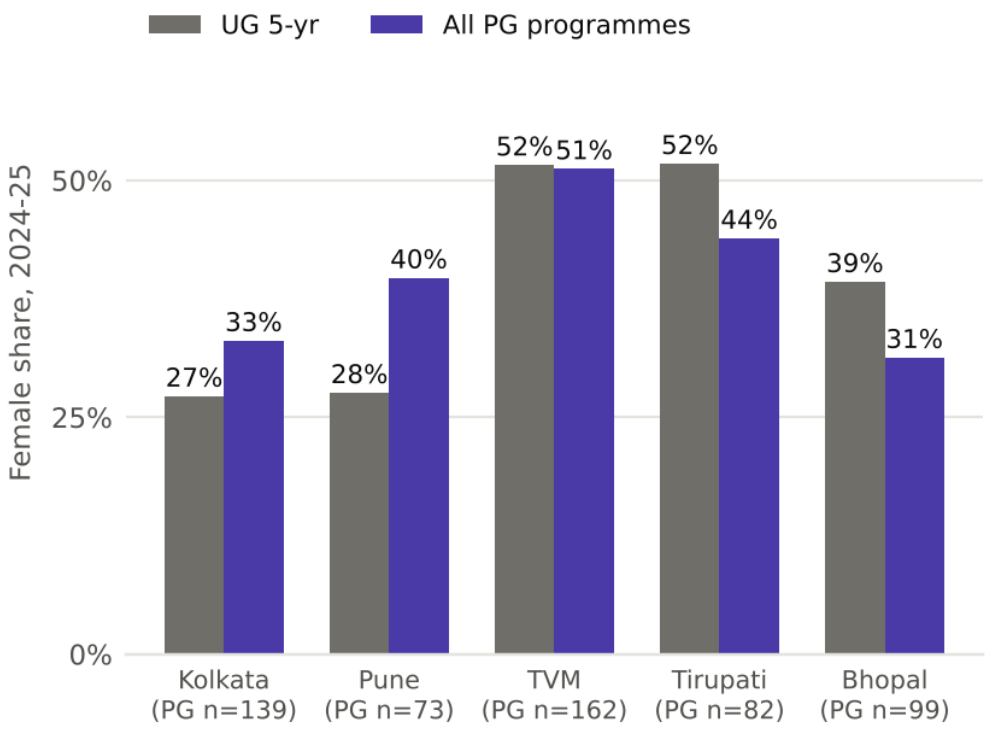


Figure 6: Female share in the UG five-year programme and in all postgraduate programmes, 2024-25. PG n: number of postgraduate students.

5. Does the home state explain it?

One explanation we wanted to test is that institutes drawing more students from their own state enrol more women, perhaps because families are more willing to send daughters somewhere closer due to the cultural norms that prevail in the country. Across the seven institutes it does not hold: the correlation is about zero ($r = -0.06$, $n = 7$).

Tirupati takes about 7% of its students from its own state and has the highest female share (51.8%). TVM takes 49% and has 51.6%. Kolkata takes 39% and has 27.2%.

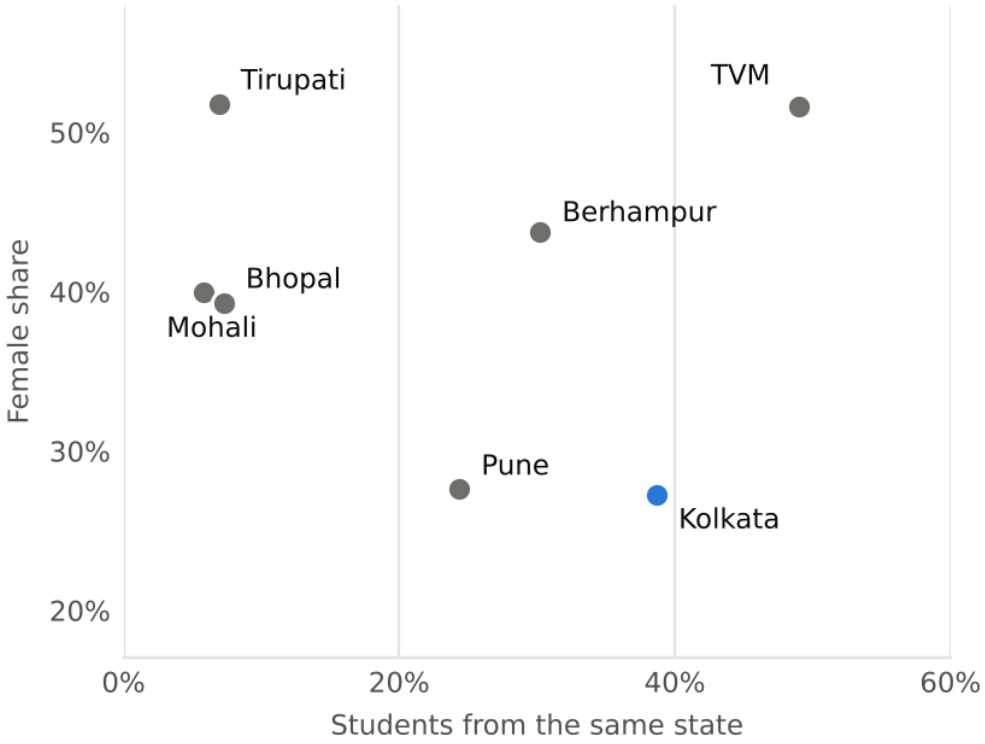


Figure 7: Students from the same state against female share, latest year. Each point is one institute. Blue: IISER Kolkata.

Over time, TVM's home-state share fell from 68% (2018-19) to 49% (2024-25) while its female share first rose and then dipped, so the within-institute pattern is mixed. With so few institutes this does not rule out an effect, but it gives no support to the explanation.

NIRF records only "within state", "outside state" and "outside country". It does not say which states students come from, so these data cannot show, for example, how many students come from North East India. Studying regional access needs state-of-origin data from the institutes.

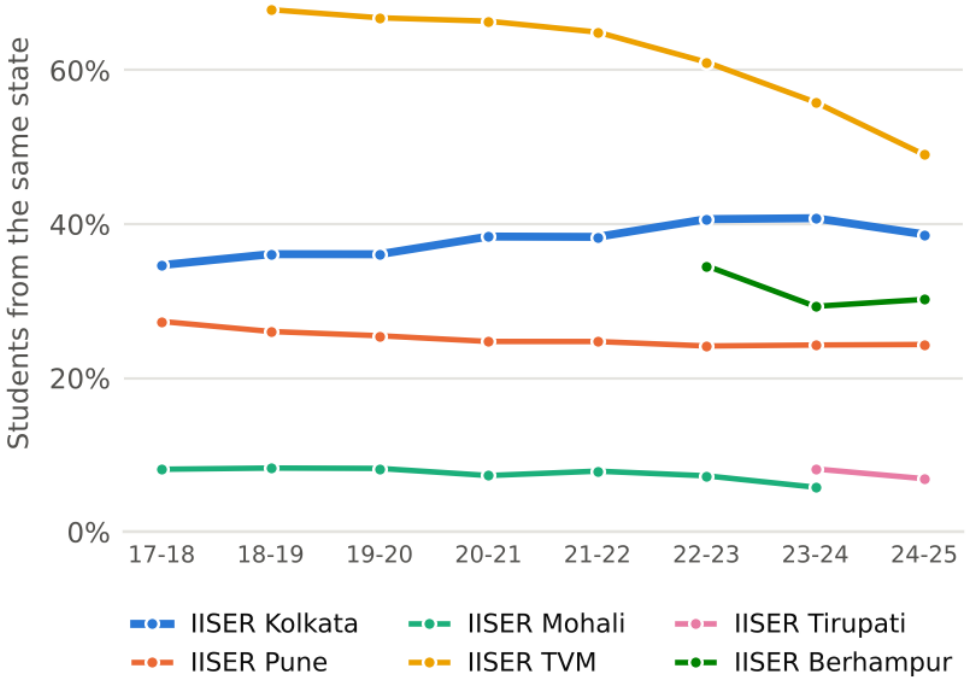


Figure 8: Students from the institute's own state over time. Share of UG five-year enrolment.

6. A contrast: reserved-category share

The NIRF table also reports students in the "socially challenged" category, which combines SC, ST and OBC. In the latest year it lies between 40% and 55% at all seven institutes, close to the nominal central quotas for these groups (15% SC, 7.5% ST and 27% OBC, together 49.5%). Gender, in contrast, varies by 25 points.

The explanation is quite easy to follow, it is common knowledge that the government in India strictly enforces seat reservation for the "socially challenged" classes. However, though the admissions follow a clear rule for categories, there is no comparable rule for gender at the IISERs as far as our research went. Hence, the difference in representation.

However, the column has limits. We could not find data that separated SC and ST students from OBC students, and nor could we find anything that showed women within a category. The share has drifted down at Pune (50% to 45%) and Mohali (53% to 48%), and the data cannot say why, including whether reporting changed. We did not use the "economically backward" column, which varies implausibly between institutes (for example 37% at Berhampur and 9 to 11% at most others).

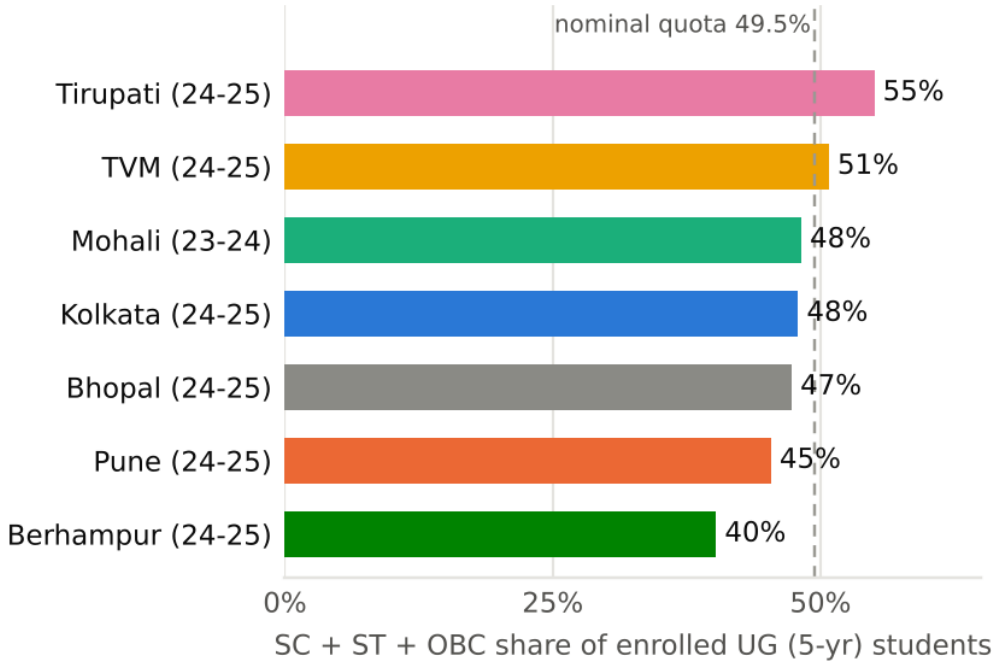


Figure 9: Share of SC, ST and OBC students (combined), latest year. Dashed line: nominal central quota total.

What it adds up to

Supported by these data

- IISER Kolkata has one of the lowest shares of women in the five-year programme among the IISERs, and the fewest women in absolute numbers. It has moved only slowly.
- Institutes differ by more than 20 points, so the share is not fixed by the programme or the subject.
- There is no sign in these files of a steady decline across the IISERs since 2017-18.
- Adding seats did not by itself raise the female share. It depended on who filled the places.

Not supported or not testable

- The home-state explanation is not supported (Section 5).
- Why institutes differ. Possible reasons include how an institute is perceived, hostel and safety arrangements, location and regional norms, the mix of admission routes, how many women apply and accept offers, and whether women leave after joining. These are questions to test, not findings and we are trying to actively find ways to test these questions.
- Whether women drop out between the undergraduate programme and the PhD. The PhD rows in these files give totals with no gender split.
- Anything about SC or ST students specifically, or about women within a category (Section 6).
- Anything about regional origin, such as North East India (Section 5).

Literature

[1] UNESCO. (2017). Cracking the code: Girls' and women's education in science, technology, engineering and mathematics (STEM). Paris: UNESCO. https://www.unesco.org/en/articles/new-unesco-report-sheds-light-gender-inequality-stem-education

[2] Wood, J. (2020). 3 things to know about women in STEM, according to the UN. World Economic Forum, citing UNESCO Institute for Statistics data. https://www.weforum.org/stories/2020/02/stem-gender-inequality-researchers-bias/

[3] Hofstra, B., Kulkarni, V. V., Munoz-Najar Galvez, S., He, B., Jurafsky, D., & McFarland, D. A. (2020). The diversity–innovation paradox in science. Proceedings of the National Academy of Sciences, 117(17), 9284–9291. https://doi.org/10.1073/pnas.1915378117

[4] National Institutional Ranking Framework (NIRF), Ministry of Education, Government of India. Data submitted by institutes, "Overall" category, 2019 to 2026. https://www.nirfindia.org. The 44 PDFs used are listed in Appendix C.

Attributions

We would like to thank Dr. Rangeet Bhattacharyya, Dr. Rahul Banerjee, Dr. Koel Das, Dr. Neelanjana Sengupta, Dr. Anindita Bhadra, Dr. Anuradha Bhat, Dr. Malancha Ta and Dr. Rumi De for their guidance, and the institutes for making their data public.

This study was designed and carried out by Dixita Adhikari of iGEM IISER Kolkata 2026.

Appendices

Appendix A. Female share (%) of enrolled UG five-year students

Institute	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25
IISER Kolkata	24.2	23.7	23.9	26.9	27.0	26.3	26.4	27.2
IISER Pune	29.8	28.3	26.2	26.6	27.6	28.6	29.2	27.6
IISER Mohali	32.5	32.9	33.7	36.7	40.3	40.9	39.9	–
IISER TVM	–	47.2	48.1	50.6	54.4	55.3	54.9	51.6
IISER Tirupati	–	–	–	–	–	–	54.0	51.8
IISER Berhampur	–	–	–	–	–	44.2	44.8	43.7
IISER Bhopal	–	–	–	–	–	–	–	39.3

A dash means no file was available for that institute and year.

Appendix B. Files used

Kolkata: NIRF 2019 to 2026 (8). Pune: 2019 to 2026 (8). Mohali: 2019 to 2025 (7). TVM: 2020 to 2026 (7). Berhampur: 2024 to 2026 (3). Tirupati: 2025 and 2026 (2). Bhopal: 2026 (1). IISc Bangalore, reference only: 2018 and 2020 to 2026 (8). Total 44.